consultation. If you're familiar with TLS (Transport Layer Security), you'll know that it creates an encrypted web channel between Alice's browser and the server. The content of Alice's communication is inside encrypted packets going back and forth. All the metadata, including the URL path, is also encrypted with the exception of the IP addresses

s of Alice and the server, along with the domain name of the site she's visiting. These have to be clear text to properly route the packets.

Imagine an eavesdropper, Earl, listening in on the interactions between Alice and `ourvirtualhealth.com`. Because of TLS, Earl will not know the content of the correspondence—it is confidential. But because of the unencrypted metadata, he can know the IP addresses of Alice and the server as well as the domain name. Is Alice's conversation private?

One way to answer this question is to ask another: could Alice repudiate her conversation with `ourvirtualhealth.com`? That is, could Earl prove to someone else that Alice was talking to `ourvirtualhealth.com`? Earl might be able to reliably identify Alice using her IP address. For example, I've had the same static IP address from my internet provider for over a decade, so I'm sure it is possible to correlate it with other information about me. To prove it was Alice, Earl would need to convince someone else of the *authenticity* of the link between Alice and her IP address.

2.7.2 Managing tradeoffs

The example in the previous section shows the tradeoff between privacy and authenticity. When you log into your bank to check your balance, your interaction is confidential, but it is not private because the bank can prove that you are the other party to the conversation. Indeed, the bank is legally obligated by anti-money laundering laws to be able to do so.

Given this, you might be tempted to decide that privacy is a hopeless cause since almost every online conversation is done inside authenticated sessions. The administrators of the identity platforms know who they're talking to, and even in pseudonymous interactions may be able to identify you. This is the entire basis of the online ad industry. To delve deeper into this threat to communications privacy and its ramifications for society, I recommend the